

ELEC 872 Week 4 Lab

Stuart Fong 20287149, Temitayo Oduyemi 20112641

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1 Introduction

Local Binary Patterns (LBP) does not extend well to situations where the face is rotated or if one side of the face is more lit than the other. We improve on this work by using facial alignment algorithms, making LBP more robust to these changes. We detail our approach and implementation using Python and `mediapipe`. A diagram of facial regions we used is in Figure 1.

2 Lighting Correction

This is done on the observation that the main lighting discrepancy seen in the input image is that half of the face may be occluded by shadow, whereas the other half is brighter.

To correct, our solution finds two landmarks at the border of both cheeks in the image. To obtain the regions that constitutes "the cheek" on both sides, we include the following area surrounding our left and right cheek landmarks:

- Left-Cheek Landmark: Y-Direction: -10 to 20, X-Direction: +2 to +20
- Right-Cheek Landmark: Y-Direction: -10 to 20, X-Direction: -20 to -1

These pixel intensities are stored in two numpy arrays that represent the intensities for the left and right cheek, respectively. We obtain the median pixel intensities for each of these numpy arrays. This gives us a representation of the difference in lighting between the left and right sides of our image. We then multiply our original image by a gradient of pixel intensities opposite to that of our shadowed image. This normalizes the pixel intensities of our image, so that both sides have a similar intensity.

3 Rotation Correction

This normalization step corrects the fact that some image faces may be rotated away from horizontal. Here we obtain the pixels on the border of the left and

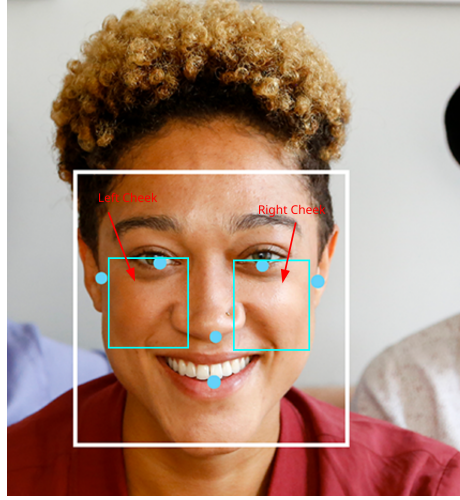


Figure 1: Face regions used by our method. We correct the intensities by getting the median of the pixel intensities of the left and right cheeks. We correct the rotation by aligning the vertical coordinates of the left and right face landmarks.

right cheek. The goal is to then realign these points so the y-coordinates between the two cheeks align.

This is done by calculating the slope between the left and right coordinates and finding the angle between them via $\arctan(\text{coordinate slope})$. We then rotate the image by the calculated angle between the slope and horizontal axis. This aligns the y-coordinates between our left and right cheek landmarks.

We then feed these transformed and normalized images to the LBP classifier.

4 Results

As opposed to improving the statistical model used to classify, the solution pre-processes the input images by transforming and normalization to reduce noise.

As shown below, the outlined image pre-processing steps greatly improve the performance of our LBP model on both the stress-test and baseline images. Resulting in a 2% increase in baseline accuracy, a 20% increase in the worst-case lighting stress-test, and a 33% increase in the worst-case rotations stress-test. Future work can include hyper-parameter tuning of the P & R parameters.

Table 1: Facial recognition results of the original and our improved LBP method under varying conditions. Our improvement is better in all scenarios.

	Original	Improved
baseline	0.975	0.995
uniform brightness	0.975	0.975
lighting ramp 0.2	0.960	0.990
lighting ramp 0.4	0.945	0.995
lighting ramp 0.6	0.895	0.960
lighting ramp 0.8	0.705	0.905
rotate 5 deg	0.970	0.975
rotate 10 deg	0.925	0.970
rotate 20 deg	0.630	0.960